CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

VI Semester of B. Sc. (Biotechnology/Microbiology/Biochemistry) Examination, May 2018

Course code & Name: BE604 & Fundamentals of Omics Technologies

Date: 09/05/2018 Day: Wednesday Time: 1:30 P.M. To 1:50 P.M. Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	**Choose the correct answer for the following questions.**		**10**
1.	The following DNA sequencing method does not require fluorescent dye labelling or gel electrophoresis instead it involves detection of pulses of chemiluminescence.		01
	(i) Sanger sequencing	(ii) Pyrosequencing	
	(iii) Chromosome walking	(iv) Both (i) & (ii)	
2.	The following are homologous genes within an organism encoding proteins with related but non-identical functions.		01
	(i) Orthologs	(ii) Pseudogenes	
	(iii) Paralogs	(iv) LINES	
3.	In 2D-PAGE, during isoelectric focusing on an immobilized pH gradient (IPG) strip, a protein in a pH region below its isoelectric point (pI) will migrate towards the __________.		01
	(i) Anode	(ii) Cathode	
	(iii) Stop migrating	(iv) Both (i) & (iii)	
4.	The genetic complexity in an organism is proportional to _______ of the genome.		01

	(i) Repetitive DNA content	(ii) Non-repetitive DNA content	
	(iii) Size	(iv) All of them	
5.	The following is an *In vivo* library-based method for screening binary protein-protein interactions.		01
	(i) Phage display	(ii) 2D-PAGE	
	(iii) Yeast two-hybrid system	(iv) Peptide mass fingerprinting	
6.	The following technique is used to determine the m/z value of each peptide fragment generated by a protein spot excised from the 2D-PAGE gel for identification of differentially expressed proteins.		01
	(i) X-ray crystallography	(ii) MALDI-TOF	
	(iii) NMR	(iv) All of them	
7.	The following transcriptome analysis technique consists of hybridization of fluorescently labelled cDNA obtained from the mRNA preparations of cells or tissues, with the probes spotted on a solid platform.		01
	(i) FISH mapping	(ii) SAGE	
	(iii) DNA microarrays	(iv) STS mapping	
8.	The following is the first free living organism whose entire genome was sequenced using the shotgun approach.		01
	(i) *Drosophila melanogaster*	(ii) *Saccharomyces cerevisiae*	
	(iii) *Haemophilus influenzae*	(iv) *Arabidopsis thaliana*	
9.	The following describes cultivation-independent analysis of collective microbial genomes contained in an environmental sample, using an approach based either on expression or sequencing.		01
	(i) Metabolomics	(ii) Metagenomics	
	(iii) Pharmacogenomics	(iv) Transcriptomics	
10.	The following describes the tandemly repetitive DNA present in the human genome.		01
	(i) Retroposons	(ii) Microsatellites	
	(iii) Minisatelites	(iv) Both (ii) & (iii)	

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Date: 09/05/2018 Day: Wednesday Time: 1:50 P.M. To 3:30 P.M. Maximum Marks: 25

Instructions:

1. **Section I and II must be attempted in SINGLE ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION – I		
Q - II	**Answer the following questions as directed (answer any two)**	**10**
1.	Explain in detail the construction of genomic libraries.	05
2.	Give an account of different classes of repetitive DNA present in the human genome.	05
3.	Explain the chemical reactions and methodology of pyrosequencing.	05
SECTION – II		
Q-III	**Answer the following questions as directed (answer any three)**	**15**
1.	Explain the phage display method for screening of binary protein-protein interactions.	05
2.	Explain the shotgun approach used for human genome sequence assembly.	05
3.	Explain the setup of a DNA microarray experiment to study differential gene expression in normal and diseased samples.	05
4.	Describe the 2D-PAGE and peptide mass fingerprinting methods for separation and identification of proteins in a proteome sample.	05
5.	Explain pharmacogenomics and explain the influence of genetic varition on individual responses to drugs.	05